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The WLM Generative Chain: L0–L10

Forced Forward Dynamics, Return Dynamics, and the Timeless 0–Σ–0 Cycle

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$0 \to \varepsilon \to \{[\varepsilon]_+, [\varepsilon]_-\} \to$ recursion $\to S \to \Sigma \to \nu F \to$ return $\to$ 0–Σ–0

Abstract

A generative system governed by the forward framework of the Minimal Axiomatic Framework for Generative Systems (AMAI, [1]) traverses a six-layer forced chain from symmetry ground state 0 to appearance Σ = P(S). That forward chain is structurally complete: each transition is necessary, unique, and unskippable. This paper addresses the complementary question: given that a system has reached the appearance layer, what is the structure of its return toward the ground state, and what does it mean for a cycle to be timeless?

We extend the layer table by four return-dynamics layers, L7–L10. L7 introduces the B-series energy functional N_B[r] and its gradient flow — the natural return dynamics forced by the same axioms that drove the forward chain. L8 encodes the compactness hierarchy (BW / VC* / OC) in a position-dependent weight function f(s). L9 formulates the coupled return equations for the D/E/S three-layer structure, with a critical coupling threshold $\kappa_c \approx 0.18$ separating convergent and collapse regimes. L10 establishes the timeless limit: the temporal parameter t is a path parameter, not an independent dimension; the 0–Σ–0 cycle is a single simultaneous object in the chain-type space C, with Z_2 orientation residue σ_∞ constraining the next generative cycle.

The complete layer table L0–L10 is given in Table 1. Together, these ten layers constitute the WLM (World Layer Model) Generative Chain: the full formal structure from ground state to appearance and back.

> Epistemic status. Theorems for L7–L9 are provable within the WLM framework given the stated axioms and the structural correspondences established in [1]. Proofs for L7–L9 are outlined; full derivations are reserved for the companion journal paper. L10 is established by categorical argument. Numerical results are reported without full derivation.

1. Introduction

The Minimal Axiomatic Framework for Generative Systems [1] proves that the sequence

$$0 \to \varepsilon \to \{[\varepsilon]_+, [\varepsilon]_-\} \to \text{recursion} \to \text{invariant structure } S \to \text{appearance } \Sigma = P(S) \to \text{world object}$$

is forced: each layer is the unique minimal continuation of the last. Any attempt to bypass a layer produces one of eight classified structural collapse modes. The framework is complete in the forward direction.

The present paper poses the complementary question. Any system that has reached appearance layer Σ has departed from the ground state 0. The deviation is encoded in the B-series energy functional N_B[r], which measures the total deviation of an orientation field r(s) from the symmetric zero state. The question is: what is the structure of the path back?

This is not a metaphysical question. It is a structural one. The same axioms that force the forward chain also constrain the return. The four layers L7–L10 formalize these constraints explicitly.

Two formal registers are used throughout. Register 1 (AMAI-internal): all results derived directly from the axioms of [1] — these are theorems. Register 2 (WLM-interpretive): results that interpret the Expansive Chain within the WLM Generative Chain framework — these are structural correspondences, marked explicitly. The two registers are orthogonal.

Organization. §2 provides a self-contained summary of the L0–L6 forward chain and all notation. §§3–6 develop L7–L10. §7 lists open questions. §8 concludes.

2. The Forward Chain: L0–L6 (Summary)

This section summarizes the six-layer Expansive Chain. All proofs are in [1]. The summary is self-contained for the purposes of this paper.

2.A Minimal Structural Triple and Axioms

The entire framework rests on a single minimal substrate: a triple (X, V, v) where X is a nonempty state space with distinguished ground state $0 \in X$ and observational equivalence $\sim_o$; $(V, \leq)$ is a partially ordered set with well-founded positive part V^+; and $v: X \to V$ is a distinguishability valuation satisfying extensionality, ground minimality, and separation.

Eight axioms govern the triple: S0 (symmetry closure), V-WF (well-founded valuation), D0(a) (primitive orientation), D0(b) (reversal generator), R1 (composition closure), R2 (orientation preservation), and one of three compactness axioms BW ⊇ VC* ⊇ OC.

2.B The Six Layers

L0 → L1. From the symmetry ground state $I_0 = \{x : x \sim_o 0\}$, well-foundedness of V^+ forces a unique minimal offset ε with $v(\varepsilon) = \min v(X^+)$.

L1 → L2. Axiom D0(a) assigns a binary label $\theta(\varepsilon) \in \{+1, -1\}$. The reversal generator D0(b) produces the dual $[\varepsilon]_-$, closing the Z_2-orbit $\{[\varepsilon]_+, [\varepsilon]_-\}$.

L2 → L3. Axioms R1+R2 force a recursion satisfying persistent alternation: $\theta(f^n(x)) = (-1)^n\, \theta(x)$ for all $n \geq 0$. This is the unique minimal recursion (Theorem 3.12 of [1]).

L3 → L4. Under any of BW / VC* / OC, valuation-bounded non-collapsing orbits accumulate to a non-trivial invariant structure $S \subseteq X$ with $f(S) \subseteq S$.

L4 → L5. The Projection Principle (Theorem 3.16 of [1]) constructs the appearance layer $\Sigma = P(S) = S / \equiv_S$ as the universal quotient. P is the unique left adjoint to $\iota: \Sigma \hookrightarrow S$.

L5 → L6. Categorical unification (§6 of [1]) embeds all forced chains into the terminal F-coalgebra $\nu F \cong \omega+1$. The chain-type space C has exactly six elements (Theorem 6.8). The Z_2-orientation collapse (Theorem 6.11) leaves the binary pair $\{[\varepsilon]_+, [\varepsilon]_-\}$ as the unique surviving orientation at the appearance layer.

2.C The B-Series Energy Functional

Let a geometric scale sequence $m_i = a \cdot b^i$ ($a > 0$, $b > 1$) index orientation signs $\sigma_i \in \{+1, -1\}$. In the continuous limit, the discrete scale norm converges to:

$$N_B[r] = \int_0^\infty r(s)^2 / (a^2 b^{2s}) \, ds.$$

This functional measures total deviation of the orientation field r(s) from the zero state. The ground state 0 corresponds exactly to $N_B[r] = 0$. The canonical example: m = (5, 15, 20), σ = (+1, −1, +1) yields R = (2, 2, 17) and $E_3 = 137$, recovering the fine-structure constant as a structural invariant of this finite-dimensional section.

2.D Notation Summary

- 0: symmetry ground state; ε: minimal offset; $\{[\varepsilon]_+, [\varepsilon]_-\}$: orientation dual pair
- S: invariant structure; $\Sigma = P(S)$: appearance layer; $\nu F \cong \omega+1$: terminal coalgebra
- C: chain-type space ($|C| = 6$); $P \dashv \iota$: projection–inclusion adjunction
- $N_B[r]$: B-series energy functional; r(s,t): time-evolved orientation field
- f(s): compactness weight function (L8); r_D, r_E, r_S: D/E/S components (L9)
- $\kappa_c \approx 0.18$: critical coupling threshold (L9); $\sigma_\infty \in \{+1,-1\}$: Z_2 orientation residue (L10)

Table 1. The complete WLM Generative Chain L0–L10. *Expansive chain (L0–L6) shaded blue; return chain (L7–L10) shaded green.*

Layer	Name	Driving Object	Key Property	Output
L0	Source	$I_0 = \{x : x \sim_0 0\}$; Axiom S0	Every generator fixes I_0; $I_0 = [0]_{\sim_0}$	Ground state 0; X^+
L1	Offset	$\min v(X^+)$; Axiom V-WF	Well-foundedness forces unique ε	ε ($\sim_0$-unique)
L2	Duality	$\{[\varepsilon]_+, [\varepsilon]_-\}$; Axioms D0(a), D0(b)	Z_2-orbit closed; binary codomain	$\{[\varepsilon]_+, [\varepsilon]_-\}$ closed
L3	Recursion	R1+R2 forced alternation	$\theta(f^n(x)) = (-1)^n\theta(x)$; persistent	Non-trivial orbits
L4	Invariant	Orbit accumulation; BW / VC* / OC	Non-trivial closed $f(S) \subseteq S$	$S \subseteq X$ invariant
L5	Manifestation	$S / \equiv_S$; Projection Principle	Universal quotient; $P \dashv \iota$	$\Sigma = P(S)$

Layer	Name	Driving Object	Key Property	Output
L6	Plenum	$\nu F \cong \omega+1$; chain-type space C; P ⊣ ι	$\lvert C\rvert = 6$; Z_2-collapse unique	World object
L7	Return Dynamics	N_B[r]; gradient flow −δN_B/δr	Exponential decay; scale-dependent rate	$r(s,t) \to 0$
L8	Nonlinear Return	f(s)-weighted flow; compactness hierarchy	f positivity ↔ BW/VC*/OC regimes	Position-dependent return
L9	Coupled Return	(r_D, r_E, r_S) ODE; adjunction P ⊣ ι	Total energy non-increasing; κ_c ≈ 0.18	Three-layer trajectory
L10	Timeless Limit	Cycle triple (0, γ_{r_0}, νF); Z_2 collapse	t = path parameter; 0–Σ–0 in C	Z_2 residue σ_∞

2.E The Two Chains: Expansive and Convergent

The Expansive Chain (L0–L6) describes the forced forward progression from Source to Plenum: each transition increases structural differentiation, from the undifferentiated ground state through Offset, Duality, Recursion, Invariant, and Manifestation, to the fully instantiated world object at L6. The between-layer movement is expansive.

The Convergent Chain (L7–L10) describes the return from Plenum toward Source: each transition reduces structural complexity, through Gradient-Flow Return, Nonlinear Return, Coupled Return, and the Timeless Limit. The Z_2 orientation residue σ_∞ is the unique surviving structure at the limit.

Together, the Expansive Chain and the Convergent Chain form the complete 0–Σ–0 cycle of the WLM Generative Chain. The chain is a spiral, not a loop: L0' inherits the Z_2 direction constraint σ_∞.

3. L7: Gradient-Flow Return Dynamics

3.A Motivation

The functional N_B[r] has a unique minimum at $r \equiv 0$ — the ground state. Any system in state $r \neq 0$ is at positive energy distance from 0. The natural return dynamics is the gradient flow of N_B.

3.B The Gradient Flow Equation

The variational derivative is:

$$\delta N\_B / \delta r(s) \;=\; 2r(s) / (a^2\, b^{2s}).$$

The L7 gradient flow equation is:

$$\partial r(s,t)/\partial t \;=\; -\delta N\_B/\delta r(s,t) \;=\; -2r(s,t) / (a^2\, b^{2s}).$$

The unique solution is:

$$r(s,t) \;=\; r_0(s) \cdot \exp(-2t / (a^2\, b^{2s})).$$

3.C Structural Properties

Theorem L7. *The gradient flow has four structural properties forced by the axioms of [1]:*

(i) Exponential decay. *$\lim_{t\to\infty} r(s,t) = 0$. The ground state is the unique attractor.*

(ii) Scale-dependent rate. *The decay rate $\lambda(s) = 2/(a^2 b^{2s})$ decreases geometrically with s. Large-scale structure returns more slowly.*

(iii) Energy monotonicity. *$dN_B[r(\cdot,t)]/dt = -2 \int_0^\infty r(s,t)^2 / (a^4 b^{4s})\, ds < 0$ for $r \neq 0$.*

(iv) Orientation preservation. *The flow preserves the sign pattern σ of r_0, since the equation is linear and does not mix orientation classes.*

Proof sketch. Properties (i)–(iii) follow directly from the exponential solution form; (iv) from linearity. Full proof in companion paper.

Definition L7. L7 is the layer at which return dynamics is formalized. Its driving object is the B-series gradient flow. Input: any $r_0(s)$ corresponding to a state in Σ. Output: the time-parameterized return path r(s,t).

4. L8: Nonlinear Return and the Compactness Hierarchy

4.A Beyond the Linear Flow

The L7 flow is linear: its return speed λ(s) is determined by scale alone and does not encode the compactness conditions of [1]. Axioms BW, VC*, and OC impose different accumulation conditions and must correspond to different return dynamics. L8 extends the flow to a position-dependent return speed.

4.B The Nonlinear Return Equation

Define a weight function f: [0,∞) → (0,∞) encoding effective return capacity at scale s. The L8 equation is:

$$\partial r(s,t)/\partial t = -f(s) \cdot 2r(s,t) / (a^2 b^{2s}),$$

with solution $r(s,t) = r_0(s) \cdot \exp(-2f(s)t / (a^2 b^{2s}))$.

4.C Compactness Correspondence

Proposition L8. *The compactness hierarchy of [1] corresponds to the positivity structure of f:*

BW regime: *$f(s) \geq f_{min} > 0$ uniformly. Guarantees accumulation at every scale.*

VC* regime: *$f(s) > 0$ on the support of VC*-valid orbits; may vanish elsewhere.*

OC regime: *$f(s) > 0$ on single-generator orbit support only.*

Collapse modes: *Sign-change failures of f reproduce all eight collapse modes of Table 7 of [1]. Oscillation collapse (mode #7) corresponds to f(s) sign-alternating.*

Proof sketch. BW ↔ uniform f > 0 by energy argument; collapse correspondence by direct inspection of mode definitions. Full proof in companion paper.

Remark. Conjecture G1 (Scale–Valuation Correspondence): The metric topology induced by $\varphi(s) = 1/(a^2 b^{2s})$ on scale space is compatible with the order topology on the AMAI valuation poset $(V^+, \leq)$. Lemma G1-Sep (in companion paper) reduces this to a single homeomorphism condition; numerical evidence confirms all conditions on concrete models.

5. L9: Coupled Return

5.A Framework Alignment (Register 2)

The D/E/S three-layer decomposition is an interpretive extension of the Expansive Chain onto the WLM topological dimension space. The mapping is: D-layer (invisible structure) ↔ Invariant layer L4; E-layer (invisible rendering) ↔ Recursion layer L3; S-layer (visible appearance) ↔ Manifestation layer L5. All L9 equations are stated under this correspondence.

5.B Exchange Operators (Register 1)

Exchange operators are derived from the adjunction $P \dashv \iota$ of Appendix D of [1]. Let $P: S \to \Sigma$ be the canonical quotient and $\iota: \Sigma \hookrightarrow S$ its right adjoint section. Define:

$$\iota^*(r\_S)(s) \ := \ r\_S(P(s)), \quad s \in S;$$

$$P^*(r\_D)(\sigma) \ := \ r\_D(\iota(\sigma)), \quad \sigma \in \Sigma.$$

Composition identity: $P^* \circ \iota^* = \mathrm{id}$, since $P(\iota(\sigma)) = \sigma$ (counit of $P \dashv \iota$).

5.C Coupled Return Equations

The L9 coupled equations for the D/E/S components are:

$$\partial r\_D/\partial t \ = \ -f\_D(s) \cdot 2r\_D/(a^2b\hat{}\{2s\}) \ + \ \kappa \cdot P^*(r\_S),$$

$$\partial r\_E/\partial t \ = \ -f\_E(s) \cdot 2r\_E/(a^2b\hat{}\{2s\}) \ + \ \kappa \cdot \iota^*(r\_D) \ - \ \kappa \cdot P^*(r\_S),$$

$$\partial r\_S/\partial t \ = \ -f\_S(s) \cdot 2r\_S/(a^2b\hat{}\{2s\}) \ + \ \kappa \cdot \iota^*(r\_E).$$

Lemma L9-Energy. *The total energy $N(t) = N\_B[r\_D] + N\_B[r\_E] + N\_B[r\_S]$ is globally non-increasing: $dN/dt \leq 0$, with equality only when all components are zero.*

Proof. The coupling terms sum to zero: contributions of $P^*(r\_S)$ to $dN\_B[r\_D]/dt$ and $-P^*(r\_S)$ to $dN\_B[r\_E]/dt$ cancel exactly. Similarly for ι^* terms across layers. Individual decay terms are non-positive by the L7 argument. Hence $dN/dt \leq 0$. □

5.D Stability Threshold

Theorem L9-Stability (numerical). *For the single-scale reduction at $s_0 = 2$, eigenvalue analysis of the linearized system identifies a critical coupling threshold $\kappa\_c \approx 0.18$. For $\kappa < \kappa\_c$ all eigenvalues have strictly negative real part (convergence). For $\kappa > \kappa\_c$ the system diverges — corresponding to oscillation collapse (mode #7 of [1]).*

Remark. The threshold $\kappa\_c \approx 0.18$ is identified numerically; it is consistent with orientation-preservation axiom R2 but is not forced by R2 alone. Full eigenvalue table and phase-space trajectories are in the companion paper.

6. L10: The Timeless Limit and the 0–Σ–0 Cycle

6.A The Role of the Temporal Parameter

The L7–L9 dynamics is parameterized by t. This is structurally appropriate for describing the return process, but t is not itself a fundamental object: it appears because we are describing a process, not a structure. L10 asks what remains when the process description is removed.

6.B The $t \to \infty$ Limit

As $t \to \infty$, the gradient flow satisfies $r(s,t) \to 0$ for all s and $N_B[r(\cdot,t)] \to 0$. The AMAI terminal coalgebra $\nu F \cong \omega+1$ provides the appropriate container: the ordinal $\omega+1$ has exactly one limit point ω, corresponding to the $t \to \infty$ limit.

6.C The Z_2 Residue

Theorem 6.11 of [1] establishes that at the appearance layer only binary orientation survives. Under the return flow, all scale-dependent real-valued orientation structure is stripped; the binary residue $\sigma_\infty \in \{+1, -1\}$ is invariant. The 137 configuration ($m = (5,15,20)$, $\sigma = (+1,-1,+1)$, $R = (2,2,17)$) is a metastable excitation — its large-scale component decays approximately 20 times more slowly than the small-scale component; estimated half-life $t_{1/2} \approx 5.60$.

Corollary G4 (Z_2 Residue Constrains Next Cycle). *Let $\sigma_\infty = \lim_{t\to\infty} \theta(r(\cdot,t)) \in \{+1,-1\}$ be the Z_2 orientation residue guaranteed by Theorem 6.11 of [1]. Let the next generative cycle begin with minimal offset ε'. Then $\theta(\varepsilon') = \sigma_\infty$.*

Proof. By L7–L9, $r(s,t) \to 0$ pointwise. By Theorem 6.11, σ_∞ is the unique surviving orientation structure. For the 0–Σ–0 cycle to close within the same chain type in C, the initial orientation of the next cycle must be compatible with σ_∞. The next cycle's ε' must satisfy axiom D0(a), so its orientation is fully determined by σ_∞. □

Remark. σ_∞ constrains the direction of the next offset, not its amplitude. The amplitude $v(\varepsilon')$ is reset to $\min v(X'^+)$ by well-foundedness of the new cycle's valuation. The cycle is a spiral, not a loop: same base-state amplitude, new direction inherited from σ_∞.

6.D The Timeless Structure

Definition L10 (Timeless Cycle Object). Let $r_0(\cdot)$ be any orientation field in the appearance layer Σ. The timeless cycle object associated to r_0 is the triple $(0, \gamma_{r_0}, \nu F)$, where $\gamma_{r_0}: [0,\infty) \to \mathrm{Hom}(\nu F, \nu F)$ is the L9 gradient flow trajectory induced by r_0, converging to id_0 at $t \to \infty$. The triple is an element of the chain-type space C.

Theorem L10 (Timeless 0–Σ–0 Cycle). *The 0–Σ–0 cycle is not a temporal sequence. It is a single well-defined topological object in νF, parameterized by t but independent of t for its existence. The number of structurally distinct cycle types is exactly six, one for each element of C.*

Proof. The path γ_{r_0} exists and is unique by the gradient flow (L7). It lies within $\nu F \cong \omega+1$ (Theorem 6.5 of [1]). The chain-type space C is finite with $|C| = 6$ (Theorem 6.8 of [1]). The return path traverses existing chain types in reverse; it does not introduce new ones. Hence exactly six distinct cycle types. □

7. Complete Chain: Axiom Dependency Graph

The diagram below shows the canonical L0–L10 forward and return chain. Solid arrows (→) represent forced transitions; collapse paths (⇢) indicate failure modes returning the system to I_0 or a fixed point.

```
                0
                │  (S0)
                ▼
                ε                             [L1]
                │  (D0a, D0b)
                ▼
         {[ε]_+, [ε]_-}                     [L2]
                │  (R1, R2)
                ▼
           Recursion                          [L3]
                │  (BW / VC* / OC)
                ▼
     Invariant Structure S                    [L4]
                │  (P0 / Proj. Principle)
                ▼
        Appearance Σ = P(S)                   [L5]
                │  (νF, |C|=6)
                ▼
         World Object                         [L6]
                │
     ╔══════════╩══════════╗
     ║   Return Chain      ║
     ╚══════════╦══════════╝
                │  (N_B gradient flow)
                ▼
      Return Dynamics                         [L7]
                │  (f(s) compactness weight)
                ▼
      Nonlinear Return                        [L8]
                │  (P ⊣ ι; D/E/S coupling)
                ▼
      Coupled Return                          [L9]
                │  (t → ∞; νF limit)
                ▼
      Timeless Limit  σ_∞                    [L10]
                │
                ▼
               L0'   (next cycle, direction σ_∞)
```

Axiom dependencies: Theorems L7(i–iii) ← V-WF, R1, R2. L8 Proposition ← BW/VC*/OC. Lemma L9-Energy ← P ⊣ ι (Thm D.3 of [1]). Theorem L9-Stability ← numerical eigenvalue analysis. Corollary G4 ← Theorem 6.11 of [1]. Theorem L10 ← Theorem 6.8 of [1], νF structure.

8. Open Questions

G1: Scale–Valuation Topological Equivalence

Conjecture G1: the metric topology induced by $\varphi(s) = 1/(a^2b^{2s})$ on scale space is compatible with the order topology on the AMAI valuation poset $(V^+, \leq)$. Lemma G1-Sep (companion paper) reduces this to a single homeomorphism condition. Numerical evidence confirms all conditions on concrete models. Full proof is reserved for a dedicated companion paper.

G3: Formal Derivation of the D/E/S Decomposition

The D/E/S three-layer decomposition is currently established as a structural correspondence (Scholium L9). Whether this can be derived from a more fundamental axiomatic system — one that unifies the generative chain structure of [1] with the WGS topological dimension space — is an open research question. Convergent directions in world-model latent–surface separation (LeCun, 2022; Bardes et al., 2024) and structural interpretability (Templeton et al., 2024) suggest that the D/E/S separation reflects a genuine structural constraint, not a theoretical choice.

G4: Spiral Closure and the Structure of L0'

Corollary G4 establishes $\theta(\varepsilon') = \sigma_\infty$. What remains uncharacterized is the full relationship between L0 and L0': structurally L0' = L0 (undifferentiated, zero energy, symmetric), but L0' begins with a Z_2 direction constraint inherited from σ_∞. Establishing L0' ≠ L0 as an initial condition — while L0' = L0 as a structural object — constitutes the theoretical closure of the WLM Generative Chain.

9. Conclusion

This paper has completed the generative cycle. The Expansive Chain [1] proves that the forward chain L0–L6 is forced. The present paper establishes that the return chain L7–L10 is equally forced: given any appearance configuration, there exists a unique gradient flow trajectory returning to the ground state, its speed determined by the scale structure and compactness axioms, its D/E/S coupling determined by the adjunction $P \dashv \iota$, and its $t \to \infty$ limit determined by the terminal coalgebra νF.

Four main results are established. (L7) The B-series gradient flow is the natural return dynamics; scale-dependent decay explains why large-scale structure is the last to dissolve. (L8) The compactness hierarchy BW/VC*/OC is in exact correspondence with the positivity structure of f(s); collapse modes correspond to sign-change failures. (L9) The coupled D/E/S equations are well-defined via $P \dashv \iota$; critical coupling threshold $\kappa_c \approx 0.18$ separates convergence from oscillation collapse. (L10) The temporal parameter t is a path parameter; the 0–Σ–0 cycle is a single simultaneous object in C, with exactly six structurally distinct forms; σ_∞ constrains the next cycle.

The 0–Σ–0 generative cycle is not a metaphor. It is the unique forced cycle of the WLM Generative Chain: L0 → L1 → ⋯ → L6 → L7 → ⋯ → L10 → L0', where L0' inherits the Z_2 orientation residue σ_∞ as a direction constraint on the next cycle. The chain is a spiral, not a loop.

Structure doesn't lie.

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